MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 7460

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Magnetron, 250 kW (min), 5450 - 5825 Mc, mechanically tuned, pulse type, air-cooled, integral magnet, unipotential cathode.

MOUNTING POSITION: Any

PHYSICAL CHARACTERISTICS:

Weight	35 lbs. (approx)
Dimensions	See figure 1
Input connections	
Input bushing	
Output coupling	
Magnet	Note 6

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ib	ib	Pi	pi	Du	tpc	F
Unit:	mAdc	A	W	kw	-	us	Mc
Maximum:	30	30	750	750	0.0012	2.5	5825
Minimum:	-	-	-	-	-	-	5450

Parameter:	Ef	epy	tk	VSWR	Alt	Tuner Torque	Anode T	Bushing T
Unit:	V	kV	Sec	-	ft	in-oz	°C	°C
Maximum:	6.5	28	-	1.5	10,000	200	120	270
Minimum:	-	-	180	-	-	-	-	-
	Note 1							

Parameter:	Output Pressurization	Cooling
Unit:	psia	Note 8
Maximum:	35	---
Minimum:	12	---
	Note 5	

TEST CONDITIONS:

Test 1

Parameter:	Ib	Du	tpc	Ef	epy	tk	VSWR	
Unit:	mAdc	-	us	V	kV	sec	-	F1 = 5450 Mc
	8.6	.000342	0.55	5	26	180	1.1	F2 = 5660 Mc
	-	-	0.45	4.8	23	-	-	F3 = 5825 Mc

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required (Note 16)

5. Preparation for delivery

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Method or Para.	Requirement or Test	Conditions	AQL (Percent Defective)	Inspection Level or Code	Symbol	Limits		Units
						Min	Max	
	Qualification							
[illegible]	Anode-cathode capacity				C	10	17	pf
[illegible]	Temperature coefficient	Anode T=50-120°C F1, F3, Notes 4 and 9			Δ F/°C	--	0.2	Mc
	Quality conformance inspection, part 1 (Note 15)							
4003	Pressurizing	30 psia min.			---	---	---	---
- - -	Standing wave ratio	VSWR = 1.1 max except where noted			---	---	---	---
1301	Heater current	Test 1			If	4.5	5.4	A
4303	Heater-cathode warmup time	Test 1, Note 1			---	---	---	---
4304	Pulse characteristics	Test 1, Note 2			---	---	---	---
4218	Tunable frequency	F1 - F3, Note 3			---	---	---	---
4306	Pulse voltage	F1, Test 1			---	---	---	---
4307	Power output	F1, F2, F3			Po	87	---	W
4308	Spectrum measurements	F1, F2, F3, Note 10			BW	---	4	Mc
4315	Starting stability	F3, Note 14			MP	---	5	%
4315	General (RF) stability	F1, F2, F3 VSWR = 1.5:1 Note 12			MP	---	0.25	%
	Quality conformance inspection, part 2							
1031	High-frequency vibration	No voltage	6.5	II				
1031	Low-frequency vibration	No voltage	6.5	II				

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Method or Para.	Requirement or Test	Conditions	AQL (Percent Defective)	Inspection Level or Code	Symbol	Limits		Units
						Min	Max	
	Quality conformance inspection, part 2 (cont'd)							
4435	Tuning	Test 1, F1 - F3	6.5	II	Tuner turns	11	14	
4451	Shock test	G=15, Note 7	6.5	II				
4310	Pulling factor	F3, Note 11	6.5	II	ΔF	--	15	Mc
	Quality conformance inspection, part 3							
4.7	Life test provisions	Test 1, Group D Note 13						
4.7.3	Life test end points				t	125		cycles
	Power output	F1, F2, F3			Po	69	--	W
	Pulse voltage	F1, Test 1			epy	22	27	kV
	Spectrum measurements	F1, F2, F3, Note 10			BW	--	5.5	Mc
	General (RF) stability	F1, F2, F3, VSWR= 1.5:1, Note 12			MP	--	0.5	%

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NOTES:

1. The maximum value specified herein is for a non-oscillating condition. Heater surge current shall not exceed 12 amperes.

2. The pulse characteristics for Test 1 shall be as follows: tsrv = 0.1 to 0.17 us, measured between the 20 and 85 percent levels of the steepest tangent above the 50 percent level. No spike or ripple shall exceed ±5 percent of the average peak value of voltage or current. The current pulse fall time shall not exceed 0.2 us (max.) as measured between 0 and 85 percent levels. Pulse shall be applied directly to plates of oscilloscope.

3. Frequency skipping or unstable operation may be encountered at some phase positions when the mismatch occurs at the end of a "long line".

4. The temperature shall be measured at the point shown on figure 1.

5. The gas used in pressurization shall provide insulating properties at least equal to that of clean dry air at the pressure specified herein.

6. In handling and mounting the magnetron, care shall be taken to prevent demagnetization. Ferromagnetic materials shall not at any time be permitted to come closer than 8 inches from the magnet, except at the ends, where the clearance may be 3 inches minimum. Energized magnets shall not at any time be permitted to come closer than 12 inches from the tube magnet.

7. The magnetron shall be mounted on a test plate and dropped five times on each of three mutually perpendicular axes parallel to the reference planes shown on figure 1. The shock pulse shall have a duration of approximately 11 milliseconds as measured at the quarter amplitude points of the acceleration shock wave.

8. Cooling requirements, where the ambient temperature does not exceed 50°C, shall be as follows: In no case shall the temperature of the anode exceed 120°C.

Average Anode Power input watts	Air Flow c.f.m.	Back Pressure (Inches of H_2O)
0	2.5	0.06
100	7.5	.16
200	10.0	.28
300	15.0	.44
400	19.5	.63
500	24.0	.87
600	29.0	1.13

9. Measurements of temperature and frequency shall be made after conditions of thermal equilibrium have been reached.

10. Stability shall not be measured under this test. The r.f. bandwidth shall be within the limits specified herein when a VSWR of 1.5/1.0 is introduced in the load at a distance of not greater than 0.5 meter from the magnetron coupling flange, the phase being adjusted for maximum bandwidth.

11. The pulling measurement shall be made in such a manner that thermal effects do not introduce appreciable errors.

12. Not more than the percentage of pulses specified herein shall result in r.f. output pulses, each having less than 70 percent of the energy content of a normal pulse. Missing pulses shall be counted within a range of plus or minus 1 percent of the test frequency during the last 3 minutes of a test interval which shall not exceed 6 minutes. Observations shall be made for the load phase position corresponding to maximum missing pulses.

13. The VSWR shall be 1.5/1.0. The interpretation of the value of the VSWR shall be as specified in Method 4310. The standing wave introducer shall be cycled continuously through a line length

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NOTES: (Cont'd)

approximately 1/2 wavelength long at an approximate rate of 4 cycles per hour. Application of operating voltages shall be in accordance with the following cycle:

(a) Cathode preheat	3 minutes
(b) Pulse voltage	7 hours
(c) Off	57 minutes

The frequency shall be changed at the start of each running period and shall be cycled between 5450 Mc, 5525 Mc, 5675 Mc, 5750 Mc, and 5825 Mc, spending one 7-hour period at each frequency.

14. Immediately after the non-operational holding period and after a 180 second heater warm-up time, the anode voltage as specified under Test (1) shall be instantaneously applied and the magnetron missing pulse count shall not exceed 5 percent over the first minute of operation. A missing pulse is defined as an r.f. pulse whose average energy is 70 percent or less than that of a normal pulse.

15. Quality conformance inspection, part 1, shall have a combined AQL (Percent defective) of 1 percent and the inspection level shall be II.

16. The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

Preparing activity:
Navy - SH
(Project 5960-N647(NAVY))

Review activity:
Navy - SH

User activities:
Navy - AS, OS, MC, CG

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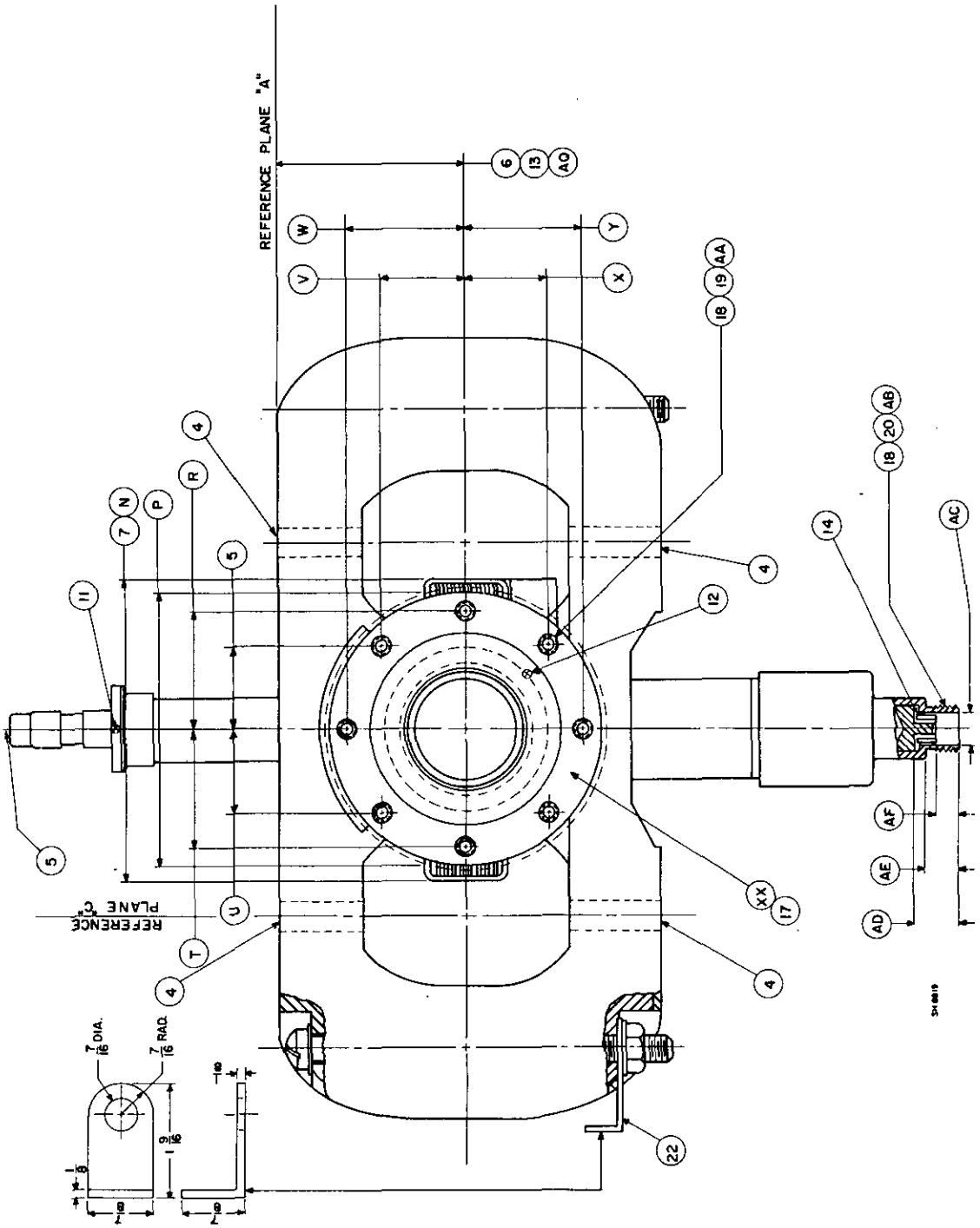


Figure 1 - Outline drawing of electron tube, type 7460.

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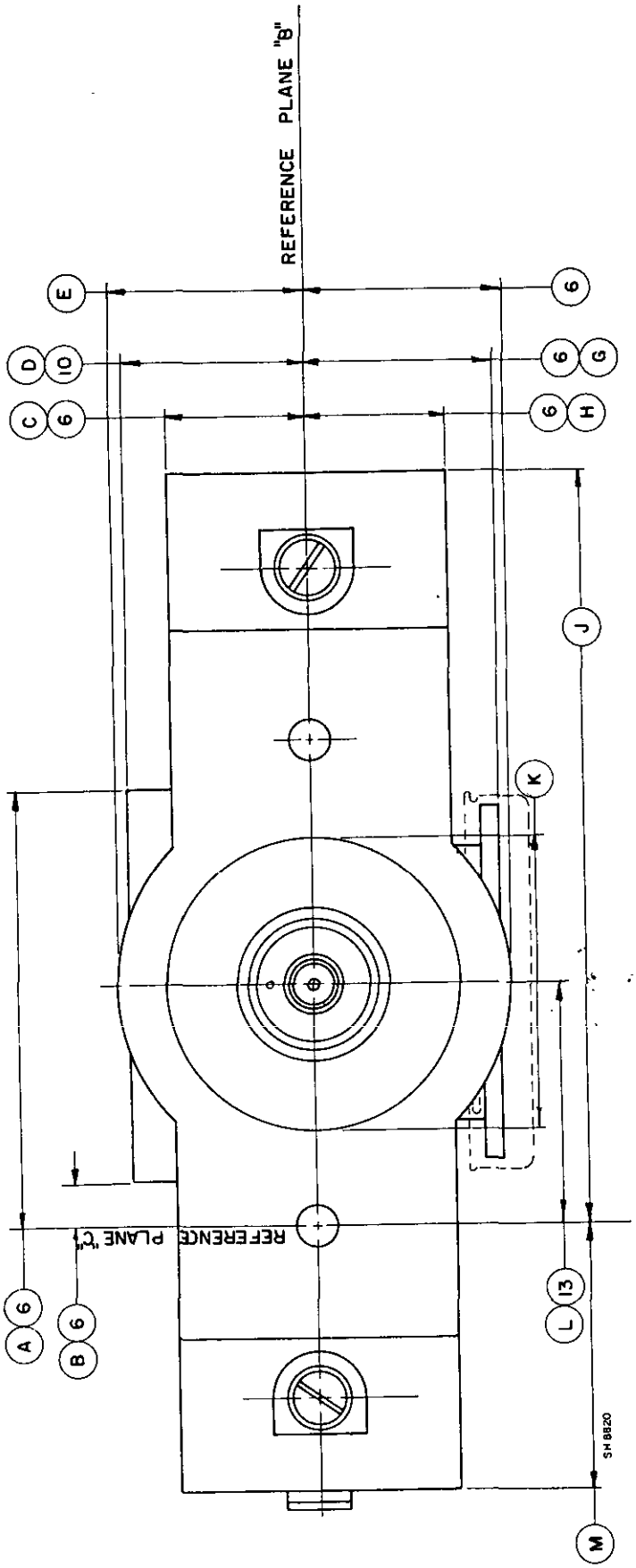


Figure 1 - Outline drawing of electron tube, type 7460 (Cont'd).

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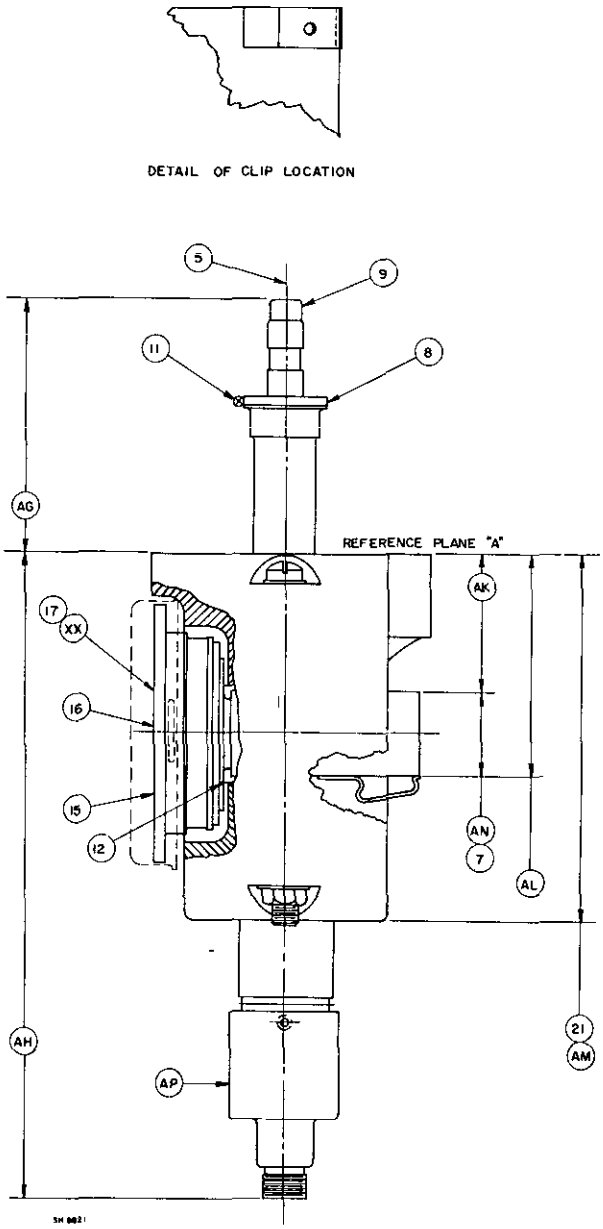


Figure 1 - Outline drawing of electron tube, type 7460 (Cont'd).

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LTR	DIMENSIONS			
	MINIMUM		MAXIMUM	
	INCHES	MM	INCHES	MM
Quality conformance inspection, part 1				
G	1.820	46.23	2.020	51.31
L	2.450	62.23	2.700	68.58
AA		10-32 NF (8 Holes)		
AB		5/8-24 NEF		
AG			3.920	99.57
AH			9.480	240.79
AQ	2.609	66.27	2.659	67.54
Quality conformance inspection part 2				
A			4.750	120.65
B	.400	10.16		
C			1.687	42.85
D			2.125	53.98
E			2.187	55.55
F			2.187	55.55
H			1.687	42.85
J			8.075	205.11
M			2.925	74.30
N	4.120	104.65	4.180	106.17
P	3.724	94.59	3.754	95.25
R	1.615	41.02	1.635	41.53
S	1.139	28.93	1.159	29.44
T	1.615	41.02	1.635	41.53
U	1.139	28.93	1.159	29.44
V	1.139	28.93	1.159	29.44
W	1.615	41.02	1.635	41.53
X	1.139	28.93	1.159	29.44
Y	1.615	41.02	1.635	41.53
AC	.435	11.05	.445	11.30
AD	.600	15.24	.654	16.61
AE	.464	11.79	.484	12.29
AF	.285	7.24	.345	8.86
AK	1.875	47.63		
AL			3.406	86.51
AM			5.437	138.10
AN	1.176	29.87	1.236	31.39
AP			1.750	44.45
Quality conformance inspection, part 3				
K	3.125 (79.38 MM) NOM.			

NOTES:

1. Reference plane "A" is defined as a plane passing along the face of the magnet as shown.
2. Reference plane "B" is defined as a plane perpendicular to reference plane "A" passing through the axes of the holes at plane "A" as shown.

Figure 1 - Outline drawing of electron tube, type 7460 (cont'd.).

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NOTES: (Cont'd)

3. Reference plane "C" is defined as a plane mutually perpendicular to planes "A" and "B" passing through the axis of the hole at plane "A" as shown.
4. These holes shall be capable of accepting a gage consisting of two 0.375 inch dia. pins, 5.500 inches long, 5.150 inches apart.
5. Parts shown on this centerline may deviate by 0.125 inch.
6. Includes angular and lateral deviation.
7. This dimension applies to size of radiator only.
8. Common heater cathode connection.
9. Heater connection.
10. This dimension is from reference plane "B" to end of radiator.
11. Cathode support temperature shall be measured at point indicated.
12. Anode temperature shall be measured at point indicated.
13. Refers to the centerline of the waveguide as determined by centers of tapped holes.
14. Spline specifications:

Pressure angle	14 degrees 30'
Pitch	48
Teeth	12
Pitch dia.	0.250

15. Output flange shall be parallel to reference plane "B" within 0.025.
16. The glass portion of the output shall not extend beyond the flange.
17. All joints in waveguide assembly shall be such that they provide an hermetic seal at surface "XX".
18. Pitch diameter shall accept a class 2 "GO" gage only.
19. Minor diameter shall not be greater than 0.166.
20. Major diameter shall not be less than 0.6145.
21. Refers to surface of magnet.
22. Support bracket: Corrosion-resistant, non-magnetic steel.

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